Gopi Krishna Veesari

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Summary

Results-driven Software Engineer with nearly 5 years of experience in full-stack development, cloud technologies, and AI-powered solutions. I build scalable backend systems using FastAPI, Django, and Flask, work with efficient databases like PostgreSQL and MySQL, and deploy applications on AWS using Docker and Kubernetes. I also have hands-on experience creating machine learning models with TensorFlow and Scikit-learn to help drive real business insights.

Technical Skills

- **Languages:** Python, JavaScript, SQL, C++, C, Java
- **Backend:** FastAPI, Flask, Django, GraphQL, RESTful APIs
- **Frontend:** React.js, HTML5, CSS3
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, EKS, CloudWatch), Docker, Kubernetes, Terraform
- **Testing & CI/CD:** Pytest, Unit test, Selenium (Python), TDD, Jenkins
- **Security:** OAuth2, JWT, SSL/TLS
- **Tools:** Git, Postman, Redis, Power BI
- **Machine Learning & AI:** Scikit-learn, TensorFlow, Pandas, NumPy, LangChain, OpenAI API, LLMs
- **Data Analysis:** Data Visualization, Data Cleaning, Data Preprocessing

Certifications

- Advanced Python Programming (NPTEL), IIT Madras
- Professional Certification in C++ Programming and Software Design, Accenture
- Specialized Certification in Machine Learning and Data Science, Skills Marathon

Experience

Capital One, USA – Software Engineer **Feb 2025 - Present**

- Designed and developed high-performance RESTful APIs using FastAPI and Python, integrating with PostgreSQL for efficient data management.
- Deployed scalable microservices on AWS EKS with Kubernetes, improving reliability.
- Implemented AI-based customer behavior prediction models using Scikit-learn, improving retention by 20%.
- Built NLP-based chatbot assistant using OpenAI API and LangChain for automated customer service.
- Integrated Redis for caching, reducing API latency by 15%.
- Developed Power BI dashboards for API usage and system health monitoring.
- Automated CI/CD pipelines via GitHub Actions, reducing release cycle times by 30%.
- Developed role-based access control and OAuth2/JWT authentication for secure API endpoints, ensuring compliance with enterprise security policies.
- Conducted load testing and performance tuning using Locust and AWS CloudWatch metrics, achieving a 25% increase in throughput.

Accenture, India – Software Engineer **Sep 2021 – Dec 2022**

- Created interactive Power BI dashboards for internal analytics and client reporting.
- Developed RESTful APIs using Flask and MySQL with optimized query handling.
- Built responsive UIs with React.js and Tailwind CSS, enhancing user engagement.
- Deployed AWS EC2 and S3 integrated solutions for high availability.
- Architected resilient Java-based microservices featuring RESTful APIs to improve system performance and reliability, increasing throughput by 30% while maintaining development principles.
- Implemented asynchronous processing with Celery and RabbitMQ, reducing API response times for bulk operations by 40%.
- Designed ETL pipelines to process and clean large datasets, enabling accurate client reporting and analytics.
- Integrated automated unit and integration testing with Pytest into CI/CD pipelines, increasing deployment reliability.
- Collaborated with cross-functional teams to migrate legacy systems to microservices, improving scalability and maintainability.

Hexaware Technologies, India – Software Engineer **Jan 2019 – Aug 2021**

- Developed microservices in Django and Python for real-time processing.
- Optimized MongoDB queries with indexing and caching.
- Trained TensorFlow models for trend prediction.
- Designed and implemented data ingestion pipelines using Kafka for streaming real-time data into MongoDB.
- Built RESTful APIs for integrating machine learning predictions into client dashboards.
- Developed scripts for data validation and transformation using Pandas and NumPy, reducing manual effort by 50%.
- Containerized deployments with Docker and orchestrated them using Kubernetes for consistent environments across stages
- Worked closely on debugging and testing backend features using Python unit tests and logging tools, helping improve overall system reliability throughout each release cycle.
- Collaborated with senior developers during agile sprints, joining code reviews and applying their feedback to consistently improve code quality and follow best practices.

Education

Master of Science in Computer Information Systems **December 2024**

Christian Brothers University, Memphis, TN

Projects:

Violent Crime Trends in Memphis, Shelby County | Power BI

- I created an interactive Power BI dashboard using public data from the Tennessee Bureau of Investigation (2020–2022) to explore violent crime trends in Memphis, Shelby County.
- Using Power BI and DAX, I built comparisons across years and months, making it easier to spot seasonal spikes in crime.
- I applied data cleaning and transformation techniques in Power Query to prepare large datasets and broke them down into clear categories like assaults, burglaries, drug cases, and weapon violations.
- To improve usability, I added interactive filters and drill-down features, allowing users to explore the data at different levels.
- This project strengthened my skills in Power BI, SQL, data visualization, data storytelling, and working with public datasets, showing how raw numbers can be turned into meaningful insights.

Fake Review Detection – Python | NLP| Machine Learning

- Developed a solution to detect fake restaurant reviews, addressing a real eCommerce challenge where misleading feedback impacts customer trust.
- Built a Python data pipeline using BeautifulSoup and Yelp API. to scrape, clean, and label review text for training datasets.
- Engineered text features with NLP techniques Count Vectorizer, TF-IDF, Hashing Vectorizer to capture meaningful language patterns.
- Created a Flask web app where users can upload reviews and test multiple ML models, making the project interactive and practical.
- Trained and evaluated Naive Bayes, Logistic Regression, Random Forest, and SVM, finding Random Forest with TF-IDF achieved the highest accuracy in detecting fake reviews.